

LAB 1 – REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

Problem Statement #38 – Dropshipping Inventory & Order Orchestrator

Name: Mutnuri Nagavalli Sri Sravya | SRN: PES1UG24CS284

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse inbound Shopify/WooCommerce order webhooks, verify supplier API inventory, and auto-dispatch supplier purchase orders.	High	Pass: Supplier PO is triggered with the customer delivery address. Fail: Order is placed for an out-of-stock supplier SKU.	Core order ingestion, inventory verification, and supplier fulfillment functionality.
FR-002	Functional	The system shall split multi-vendor orders into separate supplier-specific fulfillment orders.	High	Pass: An order containing items from multiple suppliers produces separate supplier-specific fulfillment orders. Fail: Items belonging to different suppliers remain in one supplier order.	Required to coordinate fulfillment across multiple suppliers.
FR-003	Functional	The system shall update order status based on supplier fulfillment events.	High	Pass: A supplier status event changes the corresponding order status correctly. Fail: The order status remains unchanged after a valid supplier event.	Keeps order state synchronized with fulfillment progress.
FR-004	Functional	The system shall record and track shipment tracking numbers received from suppliers.	Medium	Pass: A valid tracking number is stored against the correct fulfillment order and can be retrieved. Fail: The tracking number is missing or linked to the wrong order.	Enables shipment tracking after supplier dispatch.
FR-005	Functional	The system shall allow the E-Store Owner to view the order and fulfillment status.	Medium	Pass: The owner can retrieve an order and see its current fulfillment status. Fail: The correct order/status cannot be displayed.	Provides visibility into order processing and fulfillment.
NFR-001	Nonfunctional	The system shall acknowledge inbound webhooks with HTTP 200 within 200 ms.	High	Pass: Benchmarking confirms HTTP 200 acknowledgement within 200 ms under the specified test load. Fail: The measured acknowledgement exceeds 200 ms.	Prevents provider webhook retry storms and supports timely webhook handling.
NFR-002	Nonfunctional	The system shall maintain at least 99.9% monthly availability for order-processing and fulfillment-status services.	Medium	Pass: Monthly monitoring records at least 99.9% uptime. Fail: Availability falls below 99.9%.	Supports reliable order processing and fulfillment visibility.

Requirements Traceability Matrix (RTM)

Requirement	Mapped UML Use Case(s)	Status
FR-001	Receive & Process Order; Verify Supplier Inventory; Dispatch Supplier Purchase Order	Covered
FR-002	Split Multi-Vendor Order	Covered
FR-003	Update Order Status	Covered
FR-004	Track Shipment	Covered
FR-005	View Order & Fulfillment Status	Covered